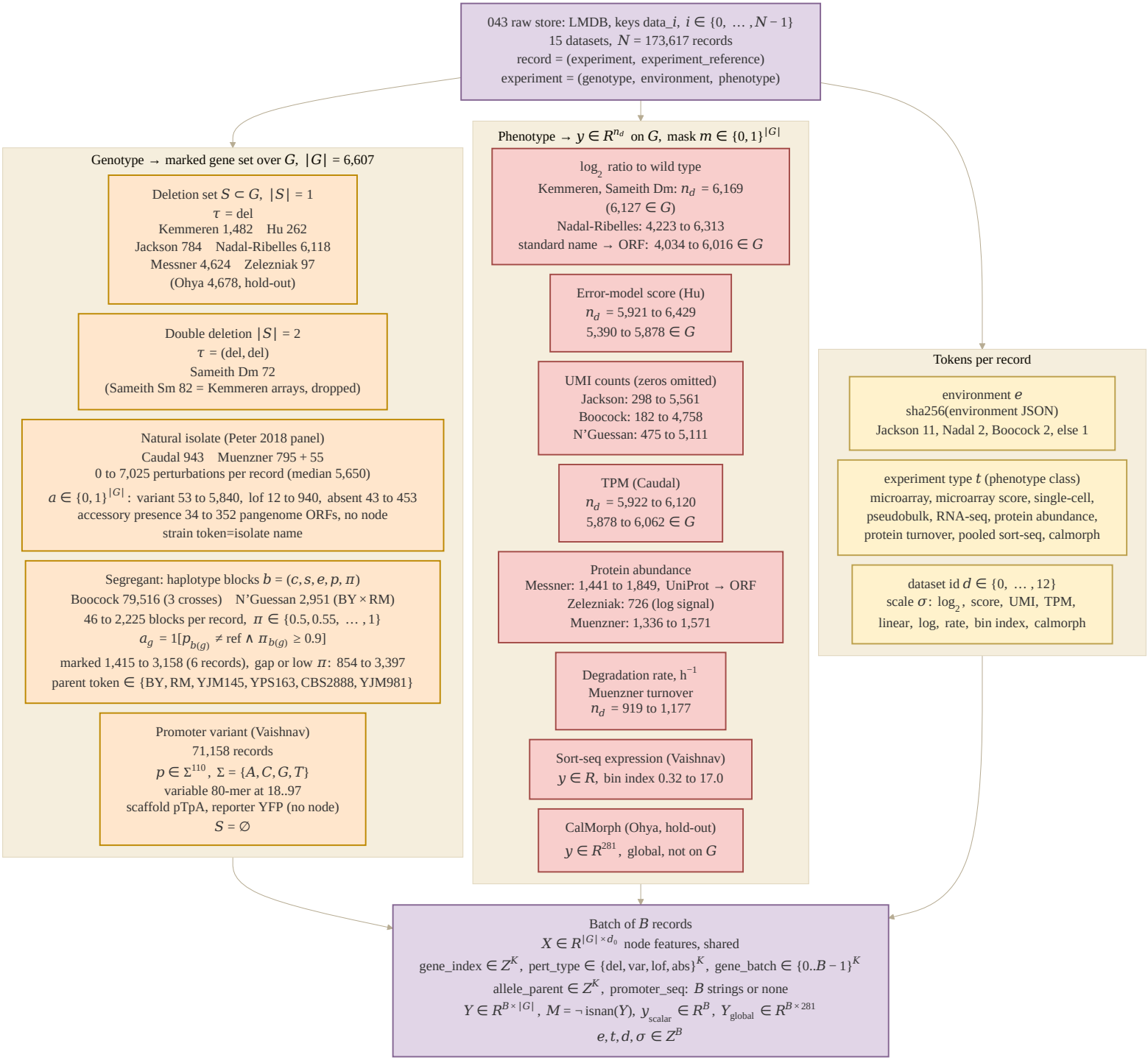

043 raw store: LMDB, keys data_i, $i \in \{0, \dots, N-1\}$
15 datasets, N = 173,617 records
record = (experiment, experiment_reference)
experiment = (genotype, environment, phenotype)
Genotype → marked gene set over G, $|G|$ = 6,607
Deletion set $S \subset G$, $|S|$ = 1
τ = del
Kemmeren 1,482 Hu 262
Jackson 784 Nadal-Ribelles 6,118
Messner 4,624 Zelezniak 97
(Ohya 4,678, hold-out)
Double deletion $|S|$ = 2
τ = (del, del)
Sameith Dm 72
(Sameith Sm 82 = Kemmeren arrays, dropped)
Natural isolate (Peter 2018 panel)
Caudal 943 Muenzner 795 + 55
0 to 7,025 perturbations per record (median 5,650)
$a \in \{0,1\}^{|G|}$: variant 53 to 5,840, lof 12 to 940, absent 43 to 453
accessory presence 34 to 352 pangenome ORFs, no node
strain token=isolate name
Segregant: haplotype blocks $b = (c, s, e, p, \pi)$
Boocock 79,516 (3 crosses) N'Guessan 2,951 (BY × RM)
46 to 2,225 blocks per record, $\pi \in \{0.5, 0.55, \dots, 1\}$
$a_g = 1[p_{b(g)} \neq \text{ref} \wedge \pi_{b(g)} \geq 0.9]$
marked 1,415 to 3,158 (6 records), gap or low π: 854 to 3,397
parent token ∈ {BY, RM, YJM145, YPS163, CBS2888, YJM981}
Promoter variant (Vaishnav)
71,158 records
$p \in \Sigma^{110}$, $\Sigma = \{A, C, G, T\}$
variable 80-mer at 18..97
scaffold pTpA, reporter YFP (no node)
$S = \varnothing$
Phenotype → $y \in R^{n_d}$ on G, mask $m \in \{0,1\}^{|G|}$
$\log_2$ ratio to wild type
Kemmeren, Sameith Dm: n_d = 6,169
(6,127 ∈ G)
Nadal-Ribelles: 4,223 to 6,313
standard name → ORF: 4,034 to 6,016 ∈ G
Error-model score (Hu)
n_d = 5,921 to 6,429
5,390 to 5,878 ∈ G
UMI counts (zeros omitted)
Jackson: 298 to 5,561
Boocock: 182 to 4,758
N'Guessan: 475 to 5,111
TPM (Caudal)
n_d = 5,922 to 6,120
5,878 to 6,062 ∈ G
Protein abundance
Messner: 1,441 to 1,849, UniProt → ORF
Zelezniak: 726 (log signal)
Muenzner: 1,336 to 1,571
Degradation rate, h^{-1}
Muenzner turnover
n_d = 919 to 1,177
Sort-seq expression (Vaishnav)
$y \in R$, bin index 0.32 to 17.0
CalMorph (Ohya, hold-out)
$y \in R^{281}$, global, not on G
Tokens per record
environment e
sha256(environment JSON)
Jackson 11, Nadal 2, Boocock 2, else 1
experiment type t (phenotype class)
microarray, microarray score, single-cell,
pseudobulk, RNA-seq, protein abundance,
protein turnover, pooled sort-seq, calmorph
dataset id $d \in \{0, \dots, 12\}$
scale σ: $\log_2$, score, UMI, TPM,
linear, log, rate, bin index, calmorph
Batch of B records
$X \in R^{|G| \times d_0}$ node features, shared
gene_index $\in Z^K$, pert_type $\in \{\text{del, var, lof, abs}\}^K$, gene_batch $\in \{0..B-1\}^K$
allele_parent $\in Z^K$, promoter_seq: B strings or none
$Y \in R^{B \times |G|}$, $M = \neg\,\text{isnan}(Y)$, $y_{\text{scalar}} \in R^B$, $Y_{\text{global}} \in R^{B \times 281}$
$e, t, d, \sigma \in Z^B$